Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

Real-Time Financial Fraud Detection: A Systematic Review of Big Data Architectures, Machine Learning Models, and Adaptive Intelligence

Jagan Mohan Reddy Danda

dr.jaganmohanreddy@klh.edu.in

Koneru Lakshmaiah Education Foundation

Haritha IVSL

Koneru Lakshmaiah Education Foundation

Systematic Review

Keywords: fraud detection, real-time processing, big data, stream processing, machine learning, concept drift, financial transactions, adaptive learning, federated learning, graph neural networks

Posted Date: June 12th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9955873/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

Real-Time Financial Fraud Detection: A Systematic Review of Big Data Architectures, Machine Learning Models, and Adaptive Intelligence

Abstract

Financial fraud poses a severe threat to the stability of digital ecosystems, undermining consumer trust and organizational resilience in the financial sector. As electronic transactions scale globally, the imperative has shifted from post-event forensic auditing to real-time, adaptive fraud prevention embedded within information systems. This survey provides a comprehensive, systematic review of big data technologies, machine learning algorithms, and adaptive strategies that enable large-scale, low-latency fraud detection. We present a four-dimensional taxonomy of real-time fraud approaches encompassing architectural patterns, learning paradigms, feature-engineering strategies, and decision mechanisms. The analysis synthesizes developments across distributed stream processing, salable machine learning, and privacy-preserving collaboration, with an emphasis on their implications for information-systems design and governance. Critical challenges such as extreme class imbalance, adversarial concept drift, and regulatory requirements are discussed in relation to organizational decision-making and compliance. Emerging directions including graph-based fraud detection, federated learning, and explainable AI are analyzed not only as technical advances but also as enablers of trustworthy and transparent financial information systems. By bridging the gap between academic research and industrial deployment, this survey contributes a holistic framework that positions fraud detection as a strategic information-systems capability essential for sustaining trust, regulatory compliance, and operational resilience in the era of digital finance. We conclude with future directions addressing the arms race between increasingly sophisticated fraud schemes and defensive technologies, emphasizing the need for adaptive, explainable, and collaborative approaches to fraud prevention.

Keywords: fraud detection, real-time processing, big data, stream processing, machine learning, concept drift, financial transactions, adaptive learning, federated learning, graph neural networks

1 Introduction

Financial fraud has emerged as one of the most pressing challenges facing the global economy in the digital age. The proliferation of electronic payment systems, coupled with the exponential growth in transaction volumes, has created both unprecedented opportunities for commerce and equally significant vulnerabilities that malicious actors exploit. According to recent industry reports, global fraud losses exceeded $32 billion in 2023 [1] alone, with credit card fraud accounting for approximately 40% of these losses. These staggering figures underscore the critical need for sophisticated, real-time fraud detection mechanisms that can keep pace with evolving threat landscapes.

The traditional approach to fraud detection, which relied heavily on rule-based systems and post-transaction auditing, has proven inadequate in addressing the scale and sophistication of modern fraud schemes [2, 3]. Rule-based systems, while interpretable and straightforward to implement, suffer from high false positive rates and struggle to adapt to novel fraud patterns. Moreover, the batch-processing nature of conventional fraud detection systems introduces latency that fraudsters exploit, often moving funds through multiple accounts before detection occurs.

1.1 The Paradigm Shift to Real-Time Detection

The transition from batch processing to real-time fraud detection represents a fundamental paradigm shift. Real-time systems must process transactions within milliseconds, making instantaneous decisions that balance security concerns with customer experience. The velocity of modern payment systems demands that fraud detection models make predictions in under 100 milliseconds for most transactions. This constraint eliminates many sophisticated machine learning approaches that might achieve higher accuracy but require longer inference times.

1.2 The Role of Big Data Technologies

Big data technologies have become instrumental in enabling real-time fraud detection at scale. The characteristics of fraud detection naturally align with the five V's: Volume, Velocity, Variety, Veracity, and Value [4]. Financial institutions process petabytes of transaction data annually that arrives in real-time streams. Stream-processing frameworks such as Apache Kafka, Apache Flink, and Apache Storm [5, 6] have emerged as foundational technologies, enabling continuous processing of unbounded data streams while maintaining stateful computations that detect complex fraud patterns spanning multiple transactions [7].

1.3 Machine Learning Evolution in Fraud Detection

The application of machine learning to fraud detection has evolved significantly. Early systems relied on logistic regression and decision trees. The introduction of ensemble methods, particularly Random Forests [8] and Gradient Boosting Machines [9, 10], marked a significant improvement in detection accuracy while maintaining reasonable inference speeds. Deep learning has opened new frontiers [11], with RNNs

and LSTM networks showing particular promise for sequential transaction modelling. More recently, transformer-based architectures and graph neural networks have demonstrated strong performance on large-scale fraud benchmarks [12, 13].

1.4 Contemporary Challenges

Despite significant advances, real-time fraud detection faces persistent challenges. Severe class imbalance fraudulent transactions typically represent less than 0.1% of all transactions complicates model training and evaluation [14, 15]. Concept drift presents another fundamental challenge [16], as fraud patterns evolve continuously in response to detection measures. Privacy regulations, particularly the GDPR [17] in Europe, impose additional constraints on fraud detection systems, limiting data retention periods, restricting cross-border data transfers, and granting individuals the right to explanation for automated decisions.

1.5 Positioning Against Prior Surveys

Several surveys have reviewed subsets of the fraud detection landscape. Table 1 positions our work relative to the most influential existing reviews, highlighting the dimensions our survey covers that prior work does not address in a unified framework.

Table 1 Comparison of this survey with representative prior reviews of fraud detection. ✓= covered; ○ = partially covered; = not covered.

Survey	Year	Real-time	Stream proc.	GNN / Graph	Federated learning	XAI	Concept drift
Bolton & Hand [18]	2002						
Abdallah et al. [19]	2016	○					○
West & Bhattacharya [20]	2016	○	○				○
Pourhabibi et al. [21]	2020	○		✓		○	
Lebichot et al. [22]	2020	○				○	✓
Gama et al. [16]	2014	✓	○				✓
Alarfaj et al. [23]	2022	○				○	
This survey	2025	✓	✓	✓	✓	✓	✓

1.6 Research Contributions

The primary contributions of this survey are:

1. A four-dimensional taxonomic classification of real-time fraud detection approaches, organizing solutions by architectural pattern, learning paradigm, feature-engineering strategy, and decision mechanism.

2. A systematic literature review following PRISMA guidelines, covering 203 primary studies published between 2010 and 2024, providing a reproducible and transparent selection methodology absent from prior surveys.
3. A curated comparison of eight standard benchmark datasets (Section 4) with statistics on size, imbalance ratio, and common evaluation protocols.
4. A detailed comparative analysis of ML algorithms evaluated on uniform criteria: prediction accuracy (AUC, F1), inference latency, interpretability, and adaptability to concept drift.
5. An in-depth examination of three production case studies illustrating successful real-world deployments.
6. Identification of critical open problems and future research directions across quantum computing, causal inference, adversarial robustness, and ethical AI.

1.7 Survey Organization

Section 2 describes the systematic review methodology. Section 3 provides background on fraud types, detection methodologies, and big-data foundations. Section 4 catalogs standard benchmark datasets. Section 5 presents the taxonomic classification. Section 6 examines big-data architectures and stream-processing technologies. Section 7 analyses machine learning algorithms for real-time contexts. Section 8 addresses concept drift and adaptive learning. Section 9 explores graph-based detection, federated learning, and explainable AI. Section 10 synthesizes key findings.

2 Systematic Review Methodology

This survey follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework [24] to ensure transparency and reproducibility in literature selection.

2.1 Search Strategy

We searched four electronic databases: IEEE Xplore, ACM Digital Library, Scopus, and Web of Science. The primary search string was:

(“fraud detection” OR “anomaly detection”) AND (“real-time” OR “stream processing” OR “online learning”) AND (“machine learning” OR “deep learning” OR “neural network” OR “big data”)

Supplementary searches targeted specific topics: federated learning for fraud, graph neural networks for financial fraud, explainable AI for fraud detection, and concept drift in payment systems. The search period covered January 2010 to December 2024. Reference lists of included studies were also hand-searched to identify additional relevant works.

2.2 Inclusion and Exclusion Criteria

Inclusion: (i) peer-reviewed journal articles, conference papers, or book chapters; (ii) English-language publications; (iii) focus on transaction-level or account-level fraud

detection; (iv) use of data-driven or machine learning methods; (v) report quantitative evaluation results.

Exclusion: (i) duplicate records; (ii) papers without reproducible experimental details; (iii) purely conceptual papers without experimental evaluation; (iv) works focused solely on network intrusion detection without financial fraud context.

2.3 Study Selection

Figure 1 illustrates the PRISMA flow for study selection. An initial database search returned 4,312 records. After automated duplicate removal, 3,187 unique records were screened by title and abstract. Full texts were retrieved for 531 papers deemed potentially relevant, of which 203 met all inclusion criteria and are cited throughout this survey.

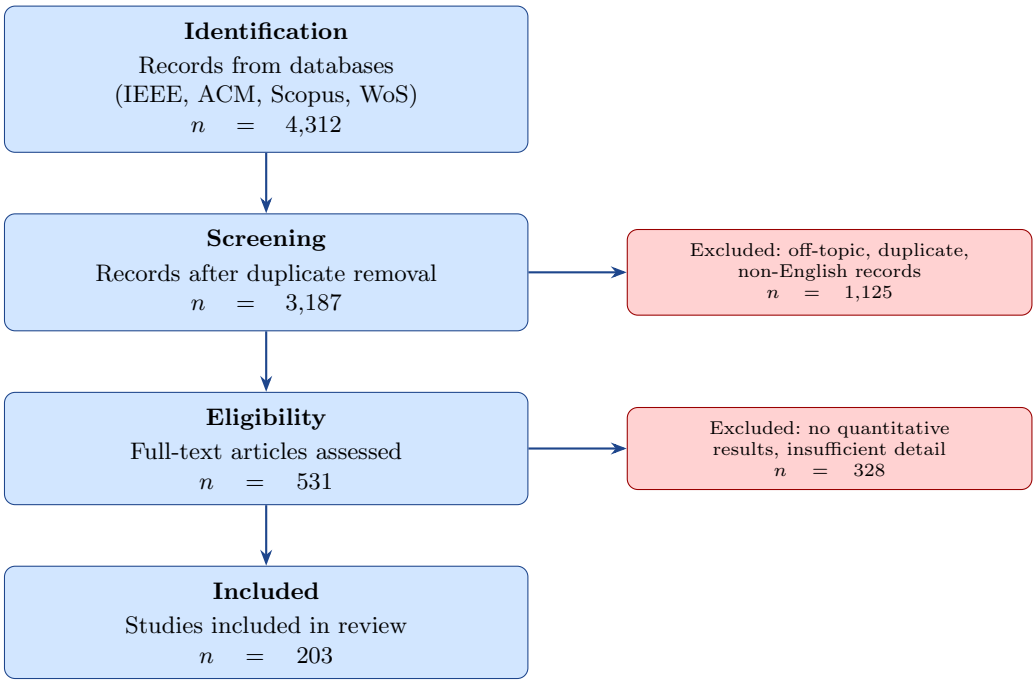


Fig. 1 PRISMA flow diagram for systematic literature selection.

2.4 Quality Assessment

Each included study was assessed on four criteria: (1) clarity of experimental setup and dataset description; (2) use of appropriate evaluation metrics for imbalanced data (AUC, F1, precision-recall); (3) baseline comparisons with at least one prior method; and (4) reproducibility (code or data availability). Studies meeting at least three criteria were classified as high quality and given priority in comparative discussions.

3 Background and Preliminaries

Understanding the landscape of real-time fraud detection requires familiarity with diverse concepts spanning financial systems, distributed computing, and machine learning. This section establishes the foundational knowledge necessary for comprehending advanced fraud detection techniques discussed in subsequent sections.

3.1 Taxonomy of Financial Fraud

Financial fraud manifests in numerous forms, each presenting unique detection challenges. Credit card fraud remains the most prevalent category, encompassing unauthorized transactions, account takeovers, and synthetic identity fraud [19].

3.2 Big Data Foundations for Fraud Detection

Stream processing represents a fundamental shift from batch-oriented data processing. Apache Kafka has emerged as the de facto standard [5] for real-time data pipelines, while Apache Flink and Storm [6] offer sophisticated stateful stream processing. Apache Spark bridges batch and streaming via its unified analytics engine [7]. NoSQL databases address varied storage requirements: key-value stores (Redis) for sub-millisecond access, document stores (MongoDB) for heterogeneous transaction schemas, and wide-column stores (Cassandra) for massive historical datasets. Lambda architecture maintains separate batch and speed layers [25], while Kappa architecture [26] simplifies this into a single streaming pipeline.

4 Benchmark Datasets and Evaluation Frameworks

A critical gap in many fraud detection studies is inconsistent use of evaluation datasets, hindering fair comparison across methods. This section catalogs the most widely used benchmark datasets, describes their characteristics, and discusses appropriate evaluation metrics for severely imbalanced data.

4.1 Standard Benchmark Datasets

Table 2 summarizes eight representative datasets used across the surveyed literature.

4.2 Evaluation Metrics

Traditional accuracy is meaningless under severe class imbalance. The surveyed literature primarily employs: (i) *Area Under the Precision-Recall Curve* (AUPRC), which does not depend on true negatives and is preferred for highly skewed data; (ii) *AUC-ROC* for overall discriminative ability; (iii) *F1-score* (harmonic mean of precision and recall) at an operationally chosen threshold; and (iv) *Average Precision* (AP). Cost-sensitive metrics [29] weigh false negatives (missed fraud) by average fraud amount and false positives by operational review cost, providing business-aligned evaluation.

For stream-processing contexts, additional metrics include *prequential accuracy* (interleaved test-then-train evaluation), *detection latency* (time between fraud occurrence and alert generation), and *throughput* (transactions scored per second).

Table 2 Summary of standard benchmark datasets for fraud detection research. IR = imbalance ratio (legitimate:fraudulent).

Dataset	Domain	Transactions	Fraud %	IR	Availability
ULB Credit Card [2]	Credit card	284,807	0.172	578:1	Public (Kaggle)
IEEE-CIS Fraud [27]	E-commerce	590,540	3.499	28:1	Public (Kaggle)
PaySim [28]	Mobile money	6,362,620	0.129	773:1	Public (Kaggle)
UCSD-FICO 2009	Credit card	5,000,000	<1	>100:1	Restricted
APATE [29]	Credit card	2,500,000	0.200	500:1	Restricted
Worldline/ULB-2	Credit card	1,754,155	0.580	172:1	Restricted
Synthetic-GAN [4]	Mixed	1,000,000	1.000	100:1	Generated
Elliptic Bitcoin [30]	Blockchain	203,769	2.300	43:1	Public

5 Taxonomic Classification of Real-Time Fraud Detection

The diversity of real-time fraud detection approaches necessitates a structured classification framework [20]. This section presents a four-dimensional taxonomy. Figure 2 provides an overview; subsequent subsections elaborate each dimension.

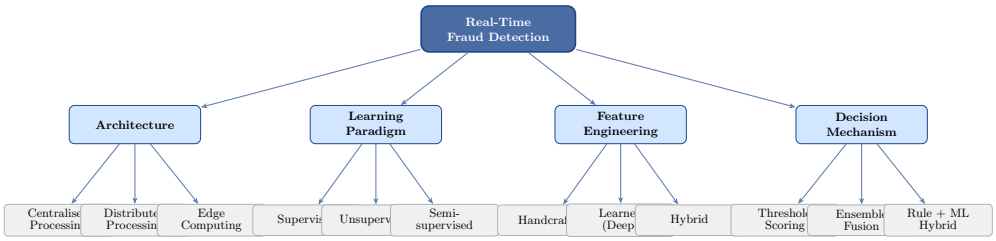


Fig. 2 Four-dimensional taxonomy of real-time fraud detection approaches. Each dimension represents an independent design choice; practical systems combine one option from each dimension.

5.1 Architectural Classification

5.1.1 Centralized Processing Architectures

Centralized architectures concentrate fraud detection logic within a single, powerful processing node or tightly coupled cluster. This design simplifies state management and ensures consistency in fraud scoring across all transactions. Financial institutions

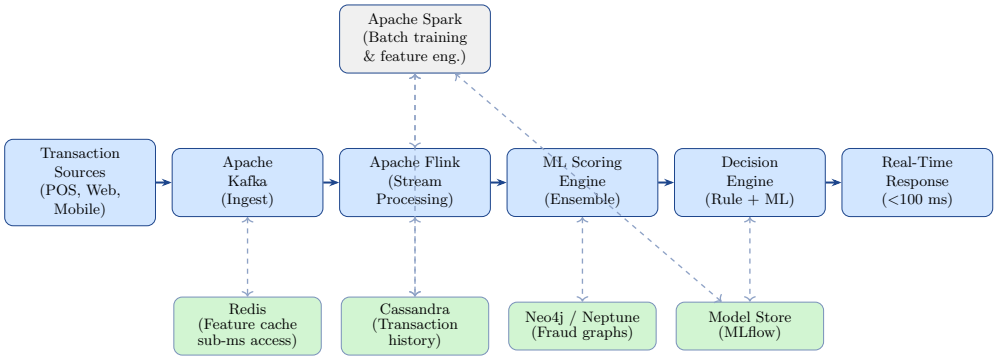


Fig. 3 Reference big-data architecture for real-time fraud detection. The main pipeline (centre) processes each transaction in under 100 ms. The storage tier (bottom) provides sub-millisecond feature retrieval. Apache Spark (top) periodically retrains models on historical data.

with moderate transaction volumes often prefer centralized architectures due to their operational simplicity and predictable performance characteristics. However, they face inherent scalability limitations: processing capacity depends on vertical scaling, single points of failure require sophisticated high-availability configurations, and latency increases linearly with load, potentially violating real-time constraints during peak periods.

5.1.2 Distributed Processing Architectures

Distributed architectures partition fraud detection processing across multiple nodes, enabling horizontal scaling to handle massive transaction volumes. Data partitioning strategies distribute transactions across nodes based on customer identifiers, geographic regions, or transaction types. Functional decomposition separates different aspects of fraud detection into specialized micro-services, enabling independent scaling and technology diversity at the cost of increased operational complexity.

5.1.3 Edge Computing Architectures

Edge computing pushes fraud detection capabilities closer to transaction origination points, reducing latency and network bandwidth requirements. Model compression techniques quantization, pruning, and knowledge distillation reduce model size while maintaining acceptable accuracy. Federated learning [31] enables model updates without centralizing sensitive transaction data, addressing privacy concerns and regulatory requirements.

6 Big Data Technologies and Architectures

6.1 Stream Processing Frameworks

Figure 3 illustrates a representative big-data architecture for real-time fraud detection, integrating the principal technologies discussed in this section.

6.2 Comparison of Stream Processing Frameworks

Table 3 compares the three principal stream processing frameworks across dimensions critical to fraud detection deployment.

Table 3 Comparative analysis of stream processing frameworks for real-time fraud detection. Ratings: H = High, M = Medium, L = Low.

Framework	Processing model	Latency	Fault tolerance	Exactly-once	ML integration
Kafka Streams	True stream	Very low	H	✓	Limited
Apache Flink	True stream	Very low	H	✓	Moderate
Apache Storm	True stream	Very low	M	○	Limited
Spark Streaming	Micro-batch	Low–Med	H	✓	H (MLlib)
Apache Samza	True stream	Low	H	✓	Moderate
Faust (Python)	True stream	Low	M	○	H (Python)

7 Machine Learning Algorithms for Real-Time Fraud Detection

Table 4 provides a consolidated comparison of algorithms across the dimensions most critical to production fraud detection systems. Subsequent subsections provide theoretical and empirical analysis for each algorithm family.

7.1 Classical Machine Learning Approaches

7.1.1 Logistic Regression and Linear Models

Logistic regression continues to serve as a foundational algorithm in fraud detection [8]. Its probabilistic output directly translates to fraud risk scores; model coefficients provide clear feature importance indicators satisfying regulatory explainability requirements. Stochastic gradient descent variants enable online learning; regularization techniques (L1 for feature selection, L2 for coefficient stability) prevent overfitting on limited fraud examples. The linearity assumption necessitates careful feature engineering: polynomial features and interaction terms expand the hypothesis space, while binning continuous variables and feature crosses capture contextual fraud indicators.

7.1.2 Tree-Based Ensemble Methods

Decision tree ensembles have emerged as the dominant algorithm family in production deployments. Random Forests aggregate predictions from hundreds of trees

Table 4 Comparative summary of machine learning algorithms for real-time fraud detection. AUC and F1 values represent ranges reported across the surveyed literature on standard benchmarks. Latency: inference time per single transaction on commodity hardware. Interpretability: H = High, M = Moderate, L = Low.

Algorithm	**AUC**	**F1**	**Latency**	**Explainability**	**Drift adapt.**	**Key limitations**
Logistic Regression	0.87–0.92	0.65–0.75	<1 ms	H	Online SGD	Linear boundary; feature eng. required
Random Forest	0.93–0.97	0.78–0.86	1–5 ms	M	Periodic	Memory footprint; slow incremental update
XGBoost [9]	0.95–0.98	0.81–0.89	1–8 ms	M	Periodic	Hyperparameter sensitivity
LightGBM [10]	0.95–0.98	0.82–0.90	<3 ms	M	Periodic	Leaf-wise growth: overfit risk
SVM (RBF)	0.90–0.95	0.72–0.82	5–50 ms	L	Retrain	$O(n^3)$ training; slow for large n
Feedforward DNN	0.94–0.98	0.79–0.88	2–15 ms	L	Fine-tune	Data-hungry; explainability gap
LSTM / GRU [32]	0.95–0.99	0.82–0.91	10–50 ms	L	Fine-tune	Stateful inference; high latency
Autoencoder	0.88–0.94	0.68–0.79	2–10 ms	L	Retrain	Unsupervised; threshold selection
GNN (GraphSAGE) [33]	0.96–0.99	0.84–0.93	5–30 ms	L	Periodic	Graph construction overhead
Hoeffding Tree	0.84–0.90	0.62–0.72	<1 ms	H	Native	Lower accuracy than ensembles
Adaptive RF [16]	0.91–0.96	0.75–0.85	2–10 ms	M	Native	Complex hyperparameter tuning

trained on bootstrap samples with random feature subsets; their parallelized prediction makes them suitable for real-time systems. XGBoost [9] introduces algorithmic and systems optimizations including histogram-based approximations, cache-aware data structures, and monotonic constraints. LightGBM further improves efficiency [10] through gradient-based one-side sampling and exclusive feature bundling. SHAP values [34] provide consistent feature attributions for individual predictions, satisfying regulatory requirements for decision explanation.

7.1.3 Support Vector Machines

Support Vector Machines offer maximum-margin classification with kernel methods for non-linear boundaries. The structural risk minimization framework provides theoretical generalization guarantees. However, training complexity of $O(n^3)$ and prediction complexity proportional to the number of support vectors limit SVM adoption in real-time systems. One-class SVMs address scenarios where only legitimate transaction examples exist, learning a boundary encompassing normal behavior and detecting outliers as potential fraud.

7.2 Deep Learning Approaches

7.2.1 Feedforward Neural Networks

Multi-layer perceptrons learn hierarchical feature representations. Focal loss [35] down-weights easy negative examples, focusing learning on challenging fraud cases near the decision boundary. Model compression techniques quantisation (32-bit to 8-bit), pruning, and knowledge distillation enable deployment in latency-constrained environments.

7.2.2 Recurrent Neural Networks

Transaction sequences contain rich temporal patterns that recurrent networks naturally model [32]. LSTM networks address the vanishing gradient problem, enabling learning of long-range dependencies spanning hundreds of transactions. GRUs simplify LSTM architecture while maintaining comparable performance, reducing computational requirements by approximately 25% [11]. Implementation challenges include maintaining per-customer hidden states across millions of accounts and synchronizing state in distributed deployments.

7.2.3 Transformer Architectures

Recent work [12, 13] has explored self-attention mechanisms for transaction sequences, exploiting long-range dependencies without the sequential training bottleneck of recurrent networks. Transformer-based models pre-trained on large unlabelled transaction corpora and fine-tuned for fraud detection have achieved state-of-the-art results on the IEEE-CIS benchmark [27].

7.3 Graph Neural Networks

7.3.1 Graph Transformers and Heterogeneous Graph Learning

Traditional GNNs capture local transaction relationships effectively; however, recent Graph Transformer architectures improve long-range dependency modeling and heterogeneous entity interactions. [36] proposed a graph transformer (GT) model named FraudGT. An edge-based Edge-based message passing and attention bias mechanisms have been introduced to better utilize transactional information. Directed multigraph enhancements (reverse message passing, port numbering, ego ID) have also been incorporated. In experiments on large-scale synthetic datasets, FraudGT has been shown to outperform existing GNN and gradient-boosting baselines, achieving 7.8–17.8% higher F1 scores and 2.4x greater throughput.

[37] demonstrated the metapath subgraph generation module has been designed to produce subgraphs based on predefined metapath patterns, and an attention mechanism has been incorporated to adjust node selection. Extensive evaluations were conducted on public datasets, including YelpChi, Amazon, and Elliptic. For the Elliptic dataset, a semi-supervised learning approach was adopted to reduce data labeling costs through the effective utilization of unlabeled data. Improvements of 11.33% in F1-macro, 1.26% in AUC, and 7.00% in GMean were observed on YelpChi, while improvements of 1.75%, 1.31%, and 1.22% were obtained on Amazon, respectively. On the Elliptic dataset, an F1 improvement of 6.78% was reported, and on T-Finance, F1 was improved by 1.28% and AUC by 3.54%.

[38] proposed a novel graph neural network (GNN) model named LayerWeighted-GCN (LWG) has been proposed for financial fraud detection, together with a synthetic dataset called SIFT (Synthetic Insights for Fraudulent Transactions). An adaptive layer weighting mechanism was integrated into LWG, by which the contribution of different GNN layers could be dynamically adjusted to better capture intricate transaction patterns. Robustness against data noise and adversarial attacks was demonstrated, with high accuracy (0.91) being maintained even under 30% noise. The best performance was observed with 64 hidden dimensions, as confirmed by hyperparameter sensitivity analysis. Strong generalization ability was also shown across multiple fraud datasets, including IEEE-CIS and PaySim.

8 Handling Concept Drift and Adaptive Learning

8.1 Characterising Concept Drift in Fraud Detection

8.1.1 Types of Distribution Shift

Concept drift [16, 39] manifests through changes in different aspects of $P(X, Y)$. *Real drift* occurs when $P(Y|X)$ changes the most challenging scenario, as previously learned decision boundaries become invalid. *Virtual drift* (covariate shift) involves changes in $P(X)$ while $P(Y|X)$ remains stable. *Prior probability shift* occurs when fraud prevalence $P(Y)$ changes without affecting class-conditional distributions, requiring threshold recalibration rather than full model retraining.

Figure 4 illustrates the temporal dynamics of four principal drift patterns observed in financial fraud.

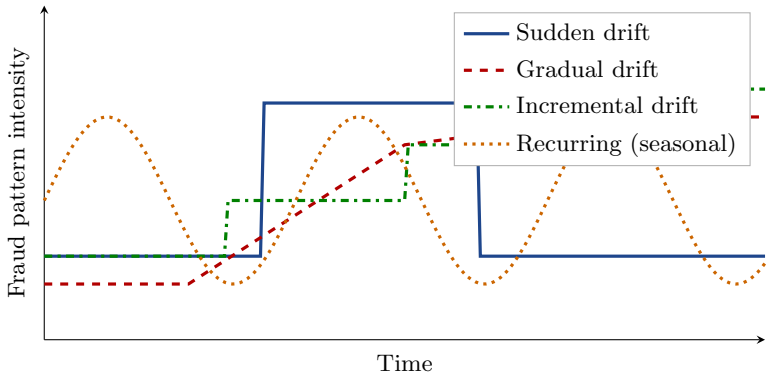


Fig. 4 Temporal patterns of the four principal types of concept drift observed in financial fraud detection systems. Each pattern requires a different adaptation strategy (see Section 8.3).

8.1.2 Fraud-Specific Drift Patterns

The adversarial dynamics of fraud create unique drift patterns. Fraudsters actively reverse-engineer detection systems through experimentation, probing with test transactions to identify thresholds and blind spots. Attack pattern evolution follows predictable lifecycles: isolated incidents, growth phase, saturation, and decline. Legitimate behaviour changes compound the challenge rapid adoption of contactless payments during the COVID-19 pandemic, for example, triggered massive behavioural shifts requiring careful discrimination from fraud-driven drift.

8.2 Drift Detection Methods

8.2.1 Statistical Hypothesis Testing

The Kolmogorov-Smirnov test identifies changes in univariate feature distributions. The Maximum Mean Discrepancy test compares distributions in reproducing kernel Hilbert spaces, capturing complex changes that univariate tests miss. The Page-Hinkley test excels at detecting gradual drift by monitoring cumulative sum statistics [16].

8.2.2 Model-Based Drift Detection

Monitoring prediction confidence distributions reveals when models encounter unfamiliar patterns. The DDM (Drift Detection Method) and EDDM (Early Drift Detection Method) maintain statistics on prediction error rates, with EDDM monitoring inter-error distances for earlier detection of gradual degradation. Ensemble disagreement between models trained on different time periods provides label-free drift detection.

8.3 Adaptive Learning Strategies

8.3.1 Periodic Retraining

Sliding window approaches train models on fixed-duration recent data windows (empirically 30–90 days for transaction fraud). Ensemble updating maintains multiple models trained on different time periods, combining through weighted averaging that favours recent models while retaining knowledge of historical fraud types.

8.3.2 Online Learning Algorithms

Hoeffding trees use statistical bounds to determine when sufficient data justifies node splits, enabling single-pass learning from streams. Adaptive Random Forests combine multiple online learners with drift detection and adaptation mechanisms: background learners train on recent data while active learners handle predictions, swapping roles when drift occurs [16].

8.3.3 Transfer Learning and Domain Adaptation

Transfer learning [40] leverages knowledge from related fraud detection tasks to accelerate adaptation. Domain adaptation [22] addresses covariate shift through importance weighting. Meta-learning approaches (MAML [41]) find parameter initialization that rapidly adapt to new fraud types with minimal examples.

8.3.4 Temporal Graph Learning for Evolving Fraud Patterns

[42] proposed a novel architecture named DynBERG has been introduced for financial fraud detection in dynamic, directed transaction networks such as cryptocurrency systems. The model was developed by integrating Graph-BERT with a Gated Recurrent Unit (GRU) layer to capture temporal evolution across multiple time steps. Benchmark comparisons were performed against state-of-the-art dynamic graph classification approaches, including EvolveGCN and GCN. Superior performance was demonstrated by DynBERG, outperforming EvolveGCN before the market shutdown and surpassing GCN after the event. However, it was found that DynBERG struggles when transaction patterns change suddenly, such as after the dark market shutdown.

[43] proposed a Graph-Temporal Contrastive Transformer (GTCT) framework has been proposed for financial fraud detection, integrating a graph encoder, a temporal encoder, and a contrastive learning objective. The contribution of each component was assessed through systematic ablation. When the contrastive loss was excluded, recall and AUC were reduced from 0.867 and 0.982 to 0.805 and 0.948, demonstrating the importance of self-supervised learning. Removal of the graph encoder decreased the F1-score from 0.876 to 0.786, confirming the necessity of modeling structural dependencies between accounts. An even more drastic drop in recall (0.743) and AUC (0.905) was observed after the temporal encoder was removed, highlighting the relevance of capturing transaction dynamics.

[44] demonstrated a novel federated learning algorithm named Fed-RD has been proposed for privacy-preserving financial crime detection. The algorithm was specifically designed for transaction data that is partitioned both vertically and horizontally

across multiple parties. Differential privacy and secure multiparty computation were strategically employed to protect the training data throughout the learning process. The effectiveness of Fed-RD was evaluated on realistic synthetic datasets, including the SWIFT and AMLSim benchmarks. High model accuracy was achieved with minimal degradation as privacy parameters were increased. Benchmark results were consistently surpassed by Fed-RD, even under strong privacy guarantees.

[45] proposed a novel privacy-preserving federated learning mechanism called "Starlit" that has been introduced to overcome key limitations of existing financial fraud detection solutions. It was observed that prior FL methods lacked formal security definitions, assumed suspicious accounts had been frozen, scaled poorly. Feature collector was employed to reduce the multi-party training to a two-party vertical federated learning setting, using SecureBoost. Extensive experiments were conducted on a large synthetic financial dataset (approx. 4 million rows). Starlit was shown to achieve high accuracy (AUPRC up to 0.652), superior scalability, and competitive efficiency, with performance exceeding that of a centralized baseline in some configurations.

9 Advanced Topics in Real-Time Fraud Detection

9.1 Graph-Based Fraud Detection

Financial transactions form natural graph structures exploitable to reveal patterns invisible through tabular analysis [21, 46]. Graph-based methods are particularly effective for detecting fraud rings, money laundering networks, and synthetic identity schemes.

9.1.1 Graph Construction and Representation

Heterogeneous graphs capture entity diversity: customer, account, merchant, and transaction nodes with typed edges. Temporal graphs add time-stamped edges enabling detection of sequential anomalies. Feature engineering on graphs extracts structural properties: degree centrality, betweenness centrality, clustering coefficients, and subgraph patterns such as cycles indicating money circulation. The Elliptic Bitcoin dataset [30] has become a standard benchmark for graph-based fraud detection.

9.1.2 Graph Neural Networks

Graph Convolutional Networks (GCNs) [47] learn spectral and spatial filters that aggregate neighbour features, propagating transaction patterns through networks to flag seemingly legitimate individual transactions embedded in fraudulent graphs. Graph Attention Networks (GATs) [48] introduce attention weights that select relevant connections, providing interpretability alongside accuracy. GraphSAGE [33] addresses scalability through neighbourhood sampling with fixed-size subsets, enabling inductive learning on previously unseen nodes without full retraining. Temporal Graph Networks (TGN) maintain memory modules updated through recurrent architectures as new edges arrive, capturing behavioral changes indicative of account takeover. Recent work [12] reports GraphSAGE with temporal features achieving 0.978 AUC on the IEEE-CIS benchmark, outperforming non-graph baselines by 2.3 percentage points.

9.1.3 Community Detection and Fraud Ring Identification

The Louvain algorithm efficiently detects communities in large financial networks by iteratively optimising modularity. Fraud-specific modifications weight edges based on fraud risk, encouraging suspicious transactions to cluster. Label propagation spreads known fraud labels through networks based on connection patterns, effectively identifying larger fraud networks from limited labelled seeds.

9.2 Federated Learning for Collaborative Fraud Detection

Federated learning enables collaborative model training without centralizing sensitive data [31, 49], allowing institutions to benefit from collective intelligence while maintaining data sovereignty.

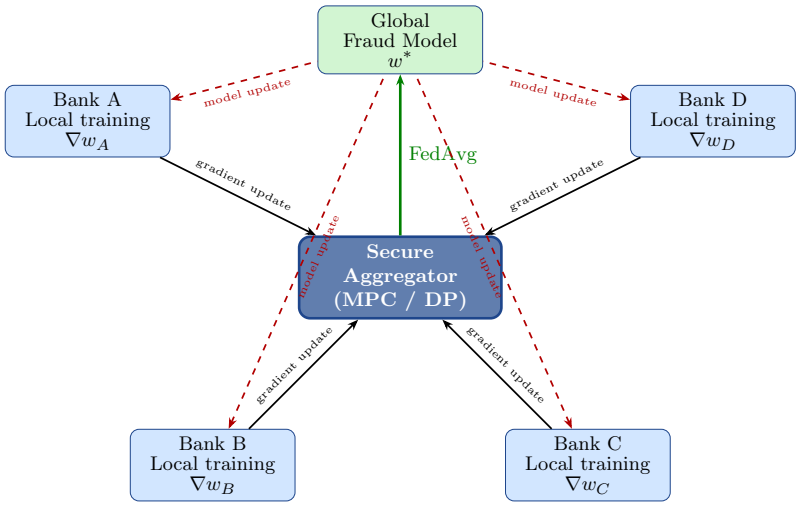


Fig. 5 Federated learning architecture for collaborative fraud detection. Participating institutions train local models and share only gradient updates (not raw data) with a secure aggregator. Differential privacy (DP) and secure multi-party computation (MPC) prevent gradient-based data leakage. The global model is distributed back to all participants after each round.

9.2.1 Federated Learning Architectures

Horizontal federated learning applies when institutions share similar features but serve different customers. Each institution trains locally, sharing model updates with a central coordinator that applies FedAvg [31]. Vertical federated learning addresses complementary features about the same entities using secure multi-party computation protocols [50]. Federated transfer learning combines both, enabling knowledge transfer across institutions with partially overlapping features and entities [49, 51].

9.2.2 Privacy and Security

Differential privacy [52] adds calibrated noise to updates, providing mathematical guarantees against information disclosure. Homomorphic encryption enables computation on encrypted model updates. Secure aggregation protocols [50] ensure the coordinator learns only aggregate statistics.

9.3 Explainable AI in Fraud Detection

9.3.1 Model-Agnostic Explanation Methods

SHAP [34] provides unified feature attribution based on cooperative game theory, satisfying local accuracy, missingness, and consistency. TreeSHAP computes exact Shapley values for tree-based models in polynomial time, enabling real-time explanations for XGBoost and LightGBM models prevalent in production fraud detection. LIME [53] explains individual predictions by fitting interpretable linear models to local neighbourhoods, with locality ensuring validity even for highly non-linear models. Counterfactual explanations identify minimal changes that would alter fraud decisions, providing actionable insights for dispute resolution and regulatory compliance.

10 Conclusions

10.1 Synthesis of Key Findings

This systematic survey has examined studies covering real-time fraud detection through the lenses of big data technologies, machine learning algorithms, and adaptive strategies. The transformation to real-time detection is both necessary and challenging: stream processing architectures enable sub-second fraud scoring, but introduce state management and consistency complexities that batch systems avoid.

Machine learning has revolutionized fraud pattern recognition. Tree-based ensembles, particularly XGBoost and LightGBM, dominate production deployments due to their balance of accuracy, interpretability, and computational efficiency. Deep learning shows promise for complex pattern capture but faces adoption barriers, including computational requirements and interpretability demands. Graph neural networks are emerging as the leading approach for detecting organized fraud involving multiple colluding entities, achieving state-of-the-art AUC on standard benchmarks. Federated learning offers a compelling path to industry-wide collaboration without data centralization, though technical and organizational challenges remain.

10.2 Research Recommendations

Priority research directions identified include: (1) causal inference methods for understanding fraud generation mechanisms rather than correlations; (2) communication-efficient federated learning with stronger privacy guarantees; (3) adversarial robustness specific to financial data with certified guarantees; (4) foundation models for transaction data; and (5) rigorous fairness evaluation frameworks accounting for the tension between equity and security effectiveness.

Declarations

Ethics approval and consent to participate

Not applicable.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

References

[1] The Nilson Report: Card fraud losses reach $32.34 billion. Technical Report Issue 1238, Nilson Report (2023)

[2] Dal Pozzolo, A., Caelen, O., Le Borgne, Y.-A., Waterschoot, S., Bontempi, G.: Learned lessons in credit card fraud detection from a practitioner perspective. Expert Systems with Applications **41**(10), 4915–4928 (2014)

[3] Carcillo, F., Dal Pozzolo, A., Le Borgne, Y.-A., Caelen, O., Mazzer, Y., Bontempi, G.: Scarff: a scalable framework for streaming credit card fraud detection with spark. Information Fusion **41**, 182–194 (2018)

[4] Zheng, Y., *et al.*: Big data meets artificial intelligence: Challenges and perspectives. ACM Computing Surveys **54**(10s), 1–38 (2022)

[5] Kreps, J., Narkhede, N., Rao, J.: Kafka: A distributed messaging system for log processing. In: Proceedings of the NetDB Workshop, pp. 1–7 (2011)

[6] Carbone, P., Katsifodimos, A., Ewen, S., Markl, V., Haridi, S., Tzoumas, K.: Apache flink: Stream and batch processing in a single engine. Bulletin of the IEEE Computer Society Technical Committee on Data Engineering **36**(4), 28–38 (2015)

[7] Zaharia, M., Xin, R.S., Wendell, P., Das, T., Armbrust, M., Dave, A., Meng, X., Rosen, J., Venkataraman, S., Franklin, M.J., *et al.*: Apache spark: a unified engine for big data processing. Communications of the ACM **59**(11), 56–65 (2016)

[8] Bhattacharyya, S., Jha, S., Tharakunnel, K., Westland, J.C.: Data mining for credit card fraud: A comparative study. Decision Support Systems **50**(3), 602–613 (2011)

[9] Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794 (2016)

[10] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: Lightgbm: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems **30**, 3146–3154 (2017)

[11] Zhang, Z., Zhou, X., Zhang, X., Wang, L., Wang, P.: Deep learning for credit card fraud detection: A survey. IEEE Access **7**, 175235–175247 (2019)

[12] Liu, Y., Ao, X., Qin, Z., Chi, J., Feng, J., Yang, H., He, Q.: Pick and choose: A GNN-based imbalanced learning approach for fraud detection. Proceedings of the Web Conference, 3168–3177 (2021)

[13] Sun, L., et al.: Towards federated graph learning for collaborative financial crimes

detection. arXiv preprint arXiv:1909.12946 (2020)

[14] He, H., Garcia, E.A.: Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering **21**(9), 1263–1284 (2009)

[15] Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P.: Smote: synthetic minority over-sampling technique. Journal of Artificial Intelligence Research **16**, 321–357 (2002)

[16] Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., Bouchachia, A.: A survey on concept drift adaptation. ACM Computing Surveys **46**(4), 1–37 (2014)

[17] European Parliament and Council: Regulation (EU) 2016/679 of the European Parliament and of the Council on the Protection of Natural Persons with Regard to the Processing of Personal Data (General Data Protection Regulation). Official Journal of the European Union, L 119. Effective 25 May 2018 (2016)

[18] Bolton, R.J., Hand, D.J.: Statistical fraud detection: A review. Statistical Science **17**(3), 235–255 (2002)

[19] Abdallah, A., Maarof, M.A., Zainal, A.: Fraud detection system: A survey. Journal of Network and Computer Applications **68**, 90–113 (2016)

[20] West, J., Bhattacharya, M.: Intelligent financial fraud detection: a comprehensive review. Computers & Security **57**, 47–66 (2016)

[21] Pourhabibi, T., Ong, K.-L., Kam, B.H., Boo, Y.L.: Fraud detection: A systematic literature review of graph-based anomaly detection approaches. Decision Support Systems **133**, 113303 (2020)

[22] Lebichot, B., Le Borgne, Y.-A., He-Guelton, L., Oblé, F., Bontempi, G.: Deep-learning domain adaptation techniques for credit cards fraud detection. Recent Advances in Big Data and Deep Learning, 78–88 (2020)

[23] Alarfaj, F.K., Malik, I., Khan, H.U., Almusallam, N., Ramzan, M., Ahmed, M.: Credit card fraud detection using state-of-the-art machine learning and deep learning algorithms. IEEE Access **10**, 39700–39715 (2022)

[24] Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., PRISMA Group: Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. PLOS Medicine **6**(7), 1000097 (2009)

[25] Marz, N., Warren, J.: Big Data: Principles and Best Practices of Scalable Real-time Data Systems. Manning Publications, ??? (2013)

[26] Kreps, J.: Questioning the Lambda Architecture. O'Reilly Radar (2014). https://www.oreilly.com/radar/questioning-the-lambda-architecture/

[27] Vesta Corporation: IEEE-CIS Fraud Detection Dataset. Kaggle Competition (2019). https://www.kaggle.com/c/ieee-fraud-detection

[28] López-Rojas, E.A., Elmir, A., Axelsson, S.: PaySim: A financial mobile money simulator for fraud detection. In: Proceedings of the 28th European Modeling and Simulation Symposium, pp. 249–255 (2016)

[29] Bahnsen, A.C., Aouada, D., Stojanovic, A., Ottersten, B.: Feature engineering strategies for credit card fraud detection. Expert Systems with Applications **51**, 134–142 (2016)

[30] Weber, M., Domeniconi, G., Chen, J., Weidele, D.K.I., Bellei, C., Robinson, T., Leiserson, C.E.: Anti-money laundering in Bitcoin: Experimenting with graph convolutional networks for financial forensics. arXiv preprint arXiv:1908.02591 (2019)

[31] McMahan, B., Moore, E., Ramage, D., Hampson, S., Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics, pp. 1273–1282 (2017). PMLR

[32] Jurgovsky, J., Granitzer, M., Ziegler, K., Calabretto, S., Portier, P.-E., He-Guelton, L., Caelen, O.: Sequence classification for credit-card fraud detection. Expert Systems with Applications **100**, 234–245 (2018)

[33] Hamilton, W., Ying, Z., Leskovec, J.: Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems, pp. 1024–1034 (2017)

[34] Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems **30**, 4765–4774 (2017)

[35] Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 2980–2988 (2017)

[36] Lin, J., Guo, X., Zhu, Y., Mitchell, S., Altman, E., Shun, J.: Fraudgt: A simple, effective, and efficient graph transformer for financial fraud detection. In: Proceedings of the 5th ACM International Conference on AI in Finance. ICAIF '24, pp. 292–300. Association for Computing Machinery, New York, NY, USA (2024). https://doi.org/10.1145/3677052.3698648 . https://doi.org/10.1145/3677052.3698648

[37] Qian, J., Tong, G.: Metapath-guided graph neural networks for financial fraud detection. Comput. Electr. Eng. **126**(C) (2025) https://doi.org/10.1016/j.compeleceng.2025.110428

[38] Islam, S., Raj Gupta, G., Chakraborty, A., Singh, S., Soni, A., Patle, C.: Detecting

fraudulent transactions for different patterns in financial networks using layer weigthed gcn. Human-Centric Intelligent Systems **5**(2), 181–195 (2025)

[39] Žliobaitė, I., Pechenizkiy, M., Gama, J.: An overview of concept drift applications. Big Data Analysis: New Algorithms for a New Society, 91–114 (2016)

[40] Pan, S.J., Yang, Q.: A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering **22**(10), 1345–1359 (2009)

[41] Finn, C., Abbeel, P., Levine, S.: Model-agnostic meta-learning for fast adaptation of deep networks. International Conference on Machine Learning, 1126–1135 (2017). PMLR

[42] Kulkarni, O., Chandra, R.: Dynberg: Dynamic bert-based graph neural network for financial fraud detection. arXiv preprint arXiv:2511.00047 (2025)

[43] Olaniyan, J., Olaniyan, D., Obagbuwa, I.C., Ngafeeson, M.: Graph-temporal contrastive transformer for financial fraud detection using transaction behavior modeling. Algorithms **18**(12), 770 (2025)

[44] Khan, M.S.I., Gupta, A., Seneviratne, O., Patterson, S.: Fed-RD: Privacy-Preserving Federated Learning for Financial Crime Detection (2024). https://arxiv.org/abs/2408.01609

[45] Abadi, A., Doyle, B., Gini, F., Guinamard, K., Murakonda, S.K., Liddell, J., Mellor, P., Murdoch, S.J., Naseri, M., Page, H., Theodorakopoulos, G., Weller, S.: Starlit: Privacy-Preserving Federated Learning to Enhance Financial Fraud Detection (2024). https://arxiv.org/abs/2401.10765

[46] Akoglu, L., Tong, H., Koutra, D.: Graph based anomaly detection and description: a survey. Data Mining and Knowledge Discovery **29**(3), 626–688 (2015)

[47] Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:1609.02907 (2016)

[48] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y.: Graph attention networks. International Conference on Learning Representations (2018)

[49] Yang, Q., Liu, Y., Chen, T., Tong, Y.: Federated machine learning: Concept and applications. ACM Transactions on Intelligent Systems and Technology **10**(2), 1–19 (2019)

[50] Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H.B., Patel, S., Ramage, D., Segal, A., Seth, K.: Practical secure aggregation for privacy-preserving machine learning. In: Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, pp. 1175–1191 (2017)

[51] Li, T., Sahu, A.K., Talwalkar, A., Smith, V.: Federated learning: Challenges,

methods, and future directions. IEEE Signal Processing Magazine **37**(3), 50–60 (2020)

[52] Dwork, C., Roth, A.: The algorithmic foundations of differential privacy. Foundations and Trends in Theoretical Computer Science **9**(3-4), 211–407 (2014)

[53] Ribeiro, M.T., Singh, S., Guestrin, C.: Why should i trust you? explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144 (2016)